JURONG JUNIOR COLLEGE
2014 JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE NAME			
CLASS		EXAM INDEX NUMBER	

CHEMISTRY

9647/02

Paper 2 Structured Questions

29 August 2014
2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, class and exam index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
The use of an approved scientific calculator is expected where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
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4	
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6	
Total	

For Examiner's Use

1 Planning (P)

The enthalpy change of reaction between carbon dioxide gas and aqueous sodium hydroxide, ΔH_r, cannot be determined directly by experiment. It is determined via an indirect approach using Hess' Law.

$$CO_2(g) + NaOH(aq) \rightarrow NaHCO_3(aq) \qquad \Delta H_r$$

The enthalpy change of the following three chemical reactions can be determined directly by experiment.

Reaction 1: $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$ $\quad \Delta H_1$
Reaction 2: $NaHCO_3(s) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l) + CO_2(g)$ $\quad \Delta H_2$
Reaction 3: $NaHCO_3(s) + aq \rightarrow NaHCO_3(aq)$ $\quad \Delta H_3$

Using these enthalpy values obtained, ΔH_r can be determined using Hess' Law.

In a preliminary examination,

- ΔH_1 was found to be approximately –57 kJ mol^{-1}.
- ΔH_2 was found to be approximately +27 kJ mol^{-1}.
- ΔH_3 was found to be approximately +18 kJ mol^{-1}.

(a) Construct an energy cycle to determine a value of ΔH_r, using the approximate values of ΔH_1, ΔH_2 and ΔH_3 obtained in the preliminary examination.

[2]

You are to design an experiment to verify the preliminary value of ΔH_2 given.

You are provided with the following information and a list of apparatus and reagents. You may also use other apparatus normally found in a school laboratory.

<u>Reagents and apparatus:</u>

- solid sodium hydrogen carbonate
- 2 mol dm^{-3} dilute hydrochloric acid
- a thermometer with 0.2 °C division
- a stopwatch
- a 200 cm^3 styrofoam cup with lid (supported in a dry beaker)

Assume that 4.2 J is required to raise the temperature of 1 cm^3 of any solution by 1 °C.

For Examiner's Use

(b) By considering the capacity of the styrofoam cup provided, state a suitable volume of hydrochloric acid you could use in the experiment.

Hence determine the mass of solid sodium hydrogen carbonate you should use in order to obtain an approximate temperature fall of 6 °C for reaction 2.

With your suggested quantities of chemicals used, prove that hydrochloric acid is used in excess in the experiment.

[2]

(c) Considering your answer in part **(b)**, write a detailed plan to verify the enthalpy change of reaction between solid sodium hydrogen carbonate and hydrochloric acid, $\Delta \boldsymbol{H_2}$.

Your plan should include:

- all essential experimental details, including precision of apparatus and quantities of chemicals used.
- a sketch of graph you expect to obtain from your experimental data to correct for heat lost to/heat gained from the surroundings.
- an outline of how you would use the experimental results to calculate $\Delta \boldsymbol{H_2}$.

In your calculations, you should adhere to the following definition of terms.

Let ***m*** g be the actual mass of solid sodium hydrogen carbonate used;
T °C be the maximum temperature change.

You must simplify the mathematical expression in your final answer.

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For Examiner's Use

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Expected graph:

Calculations to determine ΔH_2:

For Examiner's Use

[5]

(d) Concentrated hydrochloric acid is a corrosive liquid with an approximate concentration of 11 mol dm^{-3}.

Describe clearly how you would prepare 250 cm^3 of 2 mol dm^{-3} of dilute hydrochloric acid from the given concentrated acid.

State all the necessary precautions you would take to dilute the acid.

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[3]

[Total: 12]

For Examiner's Use

2 Carbides are excellent candidates for coatings for drills and other tools, as they are extremely hard and resistant to wear, corrosion and heat. Potassium and calcium carbides are commonly used in key industrial applications.

(a) **(i)** Draw the dot-and-cross diagram of potassium carbide, K_2C_2.

(ii) State the type of hybrid orbitals around the carbon atoms in the carbide anion.

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(iii) Explain whether the melting point of K_2C_2 is expected to be higher or lower than that of CaC_2.

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(iv) While calcium carbide exists as CaC_2, another ionic compound of Group II metals, beryllium carbide, exists as Be_2C instead. Suggest why this is so.

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[5]

(b) Calcium carbide is produced industrially in an electric furnace from a mixture of calcium carbonate and carbon at approximately 2000 ^{o}C. It then reacts with water to form ethyne, H–C%C–H, as one of the products.

(i) Given that the carbide anion acts like a Bronsted-Lowry base, write an equation of the reaction between calcium carbide and water.

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(ii) Ethyne is commonly used for welding metals. With the use of the *Data Booklet*, calculate the enthalpy change of combustion of ethyne. Show your working clearly.

[3]

[Total: 8]

For Examiner's Use

3 Halogens can be produced when sodium halides react with concentrated sulfuric acid via a two step-reaction with concentrated H_2SO_4 behaving as a strong acid in step 1.

Step 1: $NaX(s) + H_2SO_4(l) \rightarrow HX(g) + NaHSO_4(g)$ where X= C*l*, Br, I

Step 2: $2HX(g) + H_2SO_4(l) \rightarrow X_2(g) + SO_2(g) + 2H_2O(l)$ where X = Br, I

(a) By means of a balanced equation, show how HI is a stronger reducing agent than HC*l* and HBr.

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A student compared the reactions of sodium halides with other concentrated acids by heating NaC*l*(s), NaBr(s) and NaI(s) with concentrated H_3PO_4, H_2SeO_4 and H_6TeO_4. The observations from the reactions were summarised in the table below.

	NaC*l*(s)	NaBr(s)	NaI(s)
H_3PO_4	White fumes evolved		
H_2SeO_4	Yellowish green gas evolved	Brown gas evolved	Violet gas evolved
H_6TeO_4	No observation		

(b) Based on the observations given above, state and explain which of the three acids H_3PO_4, H_2SeO_4 or H_6TeO_4 is the

(i) weakest acid in Step 1.

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(ii) strongest oxidising agent in Step 2.

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(c) The hydrogen halides can behave as an acid by passing HX gas into water. Explain why HF(aq) is a weak acid while HC*l*(aq), HBr(aq) and HI(aq) are strong acids.

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……………………………………………………………………………………………… [1]

For Examiner's Use

3 **(d)** Concentrated hydrochloric acid is used together with nitrobenzene and tin in the synthesis of phenylamine as described below.

Preparation of phenylamine

1 Place 4 cm^3 of nitrobenzene and 10 g of tin in a 250 cm^3 two-necked round bottle flask. In one neck is placed a tap funnel and in the other is fitted with a reflux condenser.

2 Place 24 cm^3 of concentrated hydrochloric acid in the tap funnel and then add the acid a few drops at a time and shake the flask between each addition.

3 When all the acid has been added, heat the reaction mixture using a water bath for about 15 minutes.

4 Cool the flask and add aqueous sodium hydroxide until the solution is strongly alkaline. Cool the reaction mixture occasionally in ice−water bath.

5 Rearrange the apparatus for steam distillation.

(i) The first step of the synthesis involves the reaction between tin and concentrated hydrochloric acid where tin is converted to tin(II) chloride.

Write an equation to represent this reaction.

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(ii) Explain why the reaction mixture is heated using a water bath (step 3) instead of a bunsen burner.

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(iii) Suggest why the reaction mixture (step 4) has to be cooled upon adding NaOH(aq).

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(iv) If NaOH(aq) is **not** added in step 4, another organic compound (instead of phenylamine) will be formed. Give the structural formula of this compound.

For Examiner's Use

(d) (v) Phenylamine is removed from the reaction mixture by steam distillation (step 5). In steam distillation, the distilling part is infused with steam, which carries the phenylamine vapour into the condenser, where phenylamine and water is collected as a distillate.

Given that the composition of the distillate can be calculated using the following relationship:

$$\frac{\text{Amount of phenylamine}}{\text{Amount of water}} = \frac{\text{Partial pressure of phenylamine}}{\text{Partial pressure of water}}$$

and that boiling occurs when the sum of the vapour pressure equals to the atmospheric pressure,

calculate the ratio, by mass, of phenylamine to water in the distillate at an atmospheric pressure of 100 kPa when the partial pressure of phenylamine is found to be 8.0 kPa.

[6]

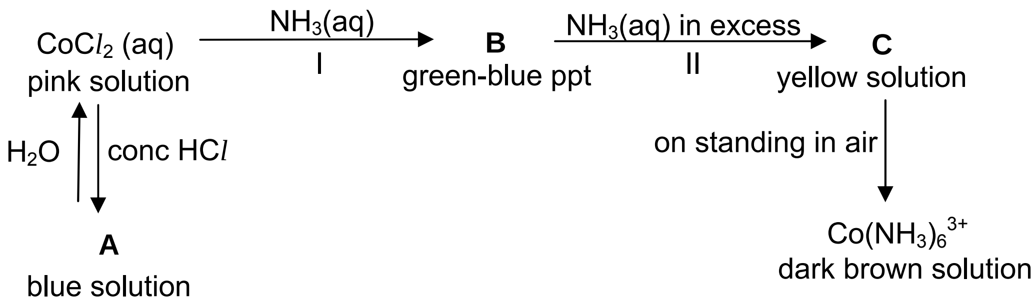
[Total: 12]

For Examiner's Use

4 Use of the *Data Booklet* is relevant to this question.

Cobalt is a typical transition metal; it is hard, fairly unreactive and is used in the production of special alloys. Cobalt also forms many coloured compounds and complexes.

(a) Aqueous cobalt(II) chloride, $CoCl_2$, is a pink solution which gives the following reactions.



(i) State the role of NH_3 in reactions I and II.

I: …………………………….…   II: …………………….…………

(ii) State the formula of compound **B**.

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(iii) What is the formula of the cation present in $CoCl_2$(aq) and the cation present in **C**?

$CoCl_2$(aq): …………………..…………… **C**: ………………..…………………

(iv) Aqueous $CoCl_2$ is chemically stable on standing in air, using relevant $E^\ominus$ values, explain why this is so.

(v) When $CoCl_2$(aq) is converted into **A**, one mole of $CoCl_2$ reacts exactly with two moles of HC*l* with no other compound formed.

Suggest the formula of the complex ion present in **A**.

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For Examiner's Use

4 (a) (vi) Dilution of **A** with water produces the original pink colour. By means of an ionic equation, explain this observation.

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(b) Cobalt and their compounds are useful catalysts. A student proposed using aqueous solution of Co^{2+} to catalyse the reaction between iodide ions, I^-, and peroxodisulfate ions, $S_2O_8^{2-}$.

By considering relevant $E^{\ominus}$ values, explain, with the aid of equations, how Co^{2+} ions can catalyse this reaction.

[4]

(c) When potassium cyanide is added to a solution containing $Co(NH_3)_6^{3+}$, a colour change of solution occurs. Suggest an explanation for this observation.

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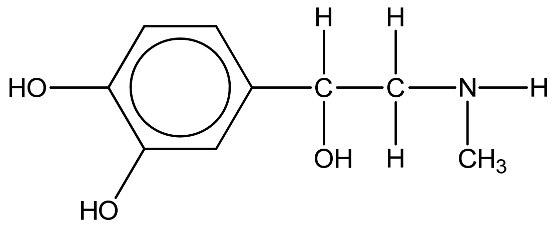
[1]

[Total: 14]

For Examiner's Use

5 Adrenaline is a hormone which acts as a stimulant when secreted directly into the blood stream. It has the following structure:



(a) **(i)** Name the functional groups present in an adrenaline molecule.

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(ii) By means of an asterisk, identify the chiral centre in the structure of adrenaline drawn above. [2]

(b) Draw the structural formula of **each** of the organic products formed when adrenaline is treated with the following reagents.

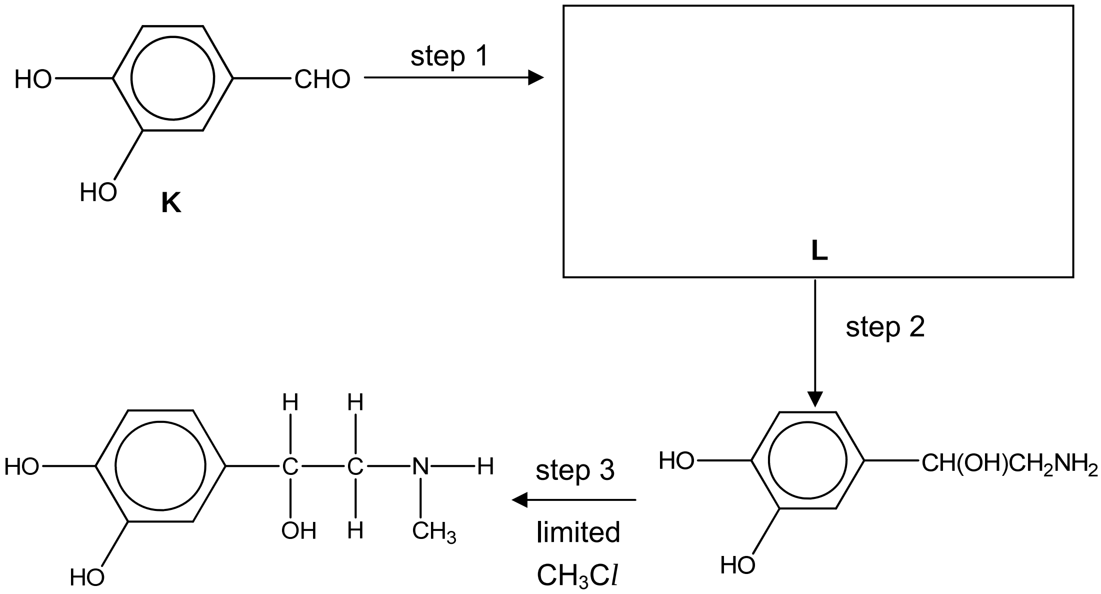
(i) an excess of bromine dissolved in a suitable solvent

(ii) hot acidified potassium dichromate(VI)

[3]

5 (c) Adrenaline can be synthsised by the following route.

For Examiner's Use



Adrenaline

(i) Draw the structural formula of **L** in the box provided.

(ii) State the reagents and conditions for step 1 and step 2.

Step 1:……………………………………………………………………………….

Step 2:……………………………………………………………………………….

(iii) Explain why step 3 is not a good method to synthesise adrenaline.

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[4]

(d) State one reagent which will react with **K** and **not** with adrenaline, and describe what will be observed.

Reagent: ………………………………………………………………………………

Observation: ……………………………………………………………………..…… [2]

[Total: 11]

For Examiner's Use

6 Esters are used commercially in perfumes and solvents. For example, 2−propyl ethanoate can be made in the laboratory by reacting propan−2−ol with compound **X.** Compound **X** gives a cream precipitate when added to $AgNO_3$(aq).

(a) **(i)** Identify the cream precipitate.

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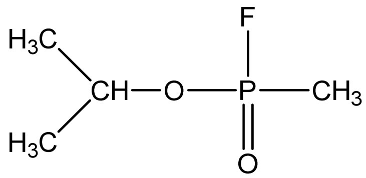
(ii) Draw the displayed formula of compound **X**.

[2]

The nerve gas *sarin,* is a phosphate ester which is extremely dangerous as it interferes with the action of a neurotransmitter in the body. It has the following structural formula.



sarin

(b) In order to destroy stockpiles of *sarin*, it is necessary to break the substance down by hydrolysis. It reacts in a similar way to an ester.

(i) State the reagents and conditions to break down an ester.

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(ii) Suggest the structural formulae of the products formed when *sarin* is hydrolysed under similar conditions stated in **b(i)**.

(iii) How would you expect the rate of hydrolysis of *sarin* to differ from that of 2−propyl ethanoate? Explain you answer.

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6 (c) *Sarin* is made by a series of chemical reactions.

The first stage is to treat methanol with phosphorus trichloride.

$$PCl_3 + 3CH_3OH \rightarrow (CH_3O)_2P(OH) + 2HCl + CH_3Cl$$

By−products produced from various side reactions can reduce the yield of $(CH_3O)_2P(OH)$ which is the desired intermediate for the next stage of the synthesis *of sarin*.

(i) Suggest a phosphorus−containing by−product that could be formed in the same reaction.

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(ii) Suggest and explain whether NCl_3 is expected to react in a similar manner with methanol.

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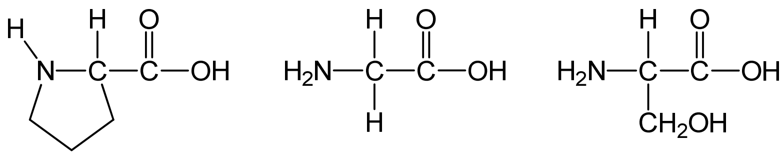
(d) The formula of three amino acids are given below.



proline (pro) glycine (gly) serine(ser)

Draw the structure of a tripeptide with the amino acid sequence pro−gly−ser showing the form in which it would exist at pH 12.

[2]

For Examiner's Use

6 **(e)** Pepsin undergoes partial hydrolysis to give the following tripeptides. Using the same 3-letter abbreviations as below, write out the amino acid sequence of the **smallest** polypeptide that could produce the following, given that the free NH_2 (N−terminal) end of the peptide is the amino acid phe.

leu−ser−glu

glu−pro−gly

gly−leu−ser

thr−glu−pro

pro−gly−ser

phe−gly−leu

……………………………………………………………………………………….. [1]

(f) Pepsin is an enzyme which aids digestion in the presence of HC*l* in the stomach. With an empty stomach, the presence of HC*l* causes gastric discomfort which can be commonly treated by medicine like *Gaviscon*® that contains $Mg(OH)_2$ as its active ingredient. However, overdosage of *Gaviscon*® may result in indigestion. Suggest and explain the impact of overdosage of *Gaviscon*® on the catalytic activity of pepsin.

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[Total: 15]